Mateo Alado

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Education

California State University, East Bay August 2022 – December 2026
Bachelor of Science in Computer Science *Hayward, CA*

- Distinctions: IEEE Xplore First Author, 2x Hackathon Winner, 3x Dean's List
- Coursework: Data Structures and Algorithms, Software Engineering, Operating Systems, AI, Statistics

Professional Experience

Lawrence Berkeley National Laboratory June 2026 – August 2026
Software Engineer Intern - Agents and Fullstack Development *Berkeley, CA*

- Engineered a REST API and LangGraph pipeline to ingest, process, and retrieve 16,000+ scientific publications
- Implemented a PostgreSQL-GraphQL API for data retrieval, reducing response time from 3+ mins to <4 seconds
- Built a React component library for knowledge graph visualization, enabling concurrent rendering of 1,000+ nodes
- Created 100+ end-to-end backend/agent tests with pytest, increasing total codebase test coverage to over 95%
- Containerized deployment of agents and services with Docker, expanding access to 450+ ALS beamline scientists
- Resolved a request concurrency bottleneck by implementing a semaphore, increasing concurrent API calls by 25%

California State University, East Bay January 2026 – May 2026
Software Research Engineer Intern - Fullstack Development *Hayward, CA*

- Developed a FastAPI service with context retrieval for data management, cutting request context size by 79.7%
- Built a script for automating data migration to a NoSQL database via cronjob, improving processing time by 60%
- Implemented an adaptive BFS traversal algorithm for data retrieval, improving Neo4j response relevance to 89%
- First-authored an IEEE Xplore paper on GenAI applications in academic research for the GAISS 2026 conference

The San Francisco Tooling Company August 2025 – April 2026
Software Engineer *Remote*

- Engineered a TypeScript agent harness with 40+ tools enabling CI/CD management and GitHub PR reviews
- Built an authentication service with Auth0 and Vercel, enabling secure credential management for 20+ engineers
- Migrated MCP tool execution to AWS Lambda for serverless hosting, reducing overall service latency by 65%
- Implemented a React UI dashboard with real-time API and tool usage statistics for users, increasing observability

Projects

Perplx - HackHayward 2026 Winner | *TypeScript, Bun, Cloudflare Workers, Cloudflare D1, MCP*

- Developed an AI coding agent with MCP tooling using the Perplexity API for context-aware software engineering
- Reduced cost per completed task to approximately $0.10 through auto compaction and prompt caching
- Built a session-sharing infrastructure using Cloudflare Workers and D1 enabling real-time collaborative sessions via shared URLs for 100+ concurrent users

Chaos Tamagotchi - Showerhacks 2026 Winner | *Swift, Python, Featherless, Xcode, AppKit*

- Ideated and built an autonomous desktop pet Tamagotchi using Swift and AppKit, incorporating a state-driven hunger engine and LLM-powered behaviors. Won $500 at Showerhacks 2026
- Implemented 5 macOs API integrations automate control over Chrome, Spotify, Finder, Contacts, and Messages
- Utilized Featherless and Mistral to generate context-aware and structured iMessage content with fallback handling

Research Management Platform | *Python, React, FastAPI, Neo4j, Supabase*

- Built and deployed a full-stack research platform using React and FastAPI for 20+ users to query and manage 100+ research datasets and papers
- Implemented persistent session context retrieval and user data storage with Supabase and Neo4j, cutting end-to-end latency by 75%+

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C, C++, Go, Swift
Frameworks: FastAPI, REST APIs, GraphQL, Node.js, Express, LangGraph, React, Next.js, PyTorch, Microservices
Data and Cloud: PostgreSQL, AWS (Lambda, DynamoDB), MongoDB, NoSQL, MySQL, Supabase, Neo4j
Tools: Docker, Git, CI/CD (GitHub Actions), Linux, UNIX, Cursor, Claude Code, LLMs, RAG, Distributed Systems